

1. The run-time on this function would be linear ($O(n)$) because it is only taking the input and checking to see if the number is even or not.
2. For this one, it would be another linear function since it is only finding a number in an array of a given length.
3. Since it multiplies every number, it would be ($O(n)$).
4. Like with the previous answer, it only multiplies the number so it is linear.
5. This one is ($O(n^2)$) because of the double "for" loops.
6. This one should be linear since it's only checking if its a prime number.
7. ($O \log(n)$) since it repeatedly divides into half.
8. Linear because each function is used once.
9.
 - a. $O(n)$
 - b. $O(n)$
 - c. $O(1)$
 - d. $O(n^3)$
 - e. $O(n^2)$
 - f. $O(n!)$
 - g. $O(n)$
 - h. $O(n^2)$